

Vivek Rallapally

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Summary

Full-stack engineer with hands-on experience building end-to-end systems using Next.js, React.js, TypeScript, FastAPI, and Python. Proficient in real-time communication, async task processing, LLM integration, and production ML pipelines — comfortable owning both the frontend and the backend.

Education

MLR Institute of Technology <i>Bachelors in Computer Science Engineering – AIML CGPA: 7.81</i>	Telangana, India 2021 – 2025
Excellencia Junior College <i>Telangana State Board of Intermediate Education: 85.2%</i>	Telangana, India 2019 – 2021
Sherwood Public School <i>Indian Certificate of Secondary Education: 73%</i>	Telangana, India 2007 – 2019

Skills Summary

Languages: Java, Python, HTML, CSS, JavaScript, TypeScript
Frameworks and Libraries: React.js, TensorFlow, scikit-learn, PyTorch, Next.js, Ollama, Flask, FastAPI, Redis, Celery, WebSockets
Tools: Android Studio, Visual Studio, Antigravity, Figma, Jupyter, Google Colab, Git, Docker
Cloud: AWS

Projects

Smart Traffic Management System Using YOLOv11 and GBML DQN <i>Reinforcement Learning</i> - Trained a GBML-DQN agent using YOLOv11 for vehicle detection in a SUMO simulation, reducing average queue length from ~55 to ~35 vehicles (~36% reduction) across 600 training episodes. - Implemented real-time vehicle counting using YOLOv11 object detection on SUMO-simulated video streams, feeding live density estimates into the DQN state space.
SED (System Execution Dashboard) <i>Agents / AI Systems / Full-Stack</i> - Built a local AI-powered laptop automation system using Python, FastAPI, Next.js, and TypeScript with real-time WebSocket communication between frontend and backend. - Integrated a local LLM (Qwen3 via Ollama) with tool calling and chaining, enabling natural language commands to execute OS-level automation including app control, system monitoring, and web search. - Implemented async task processing with Redis and Celery, supporting concurrent automation jobs with persistent conversation context across sessions.
NoPhoneDrive <i>Android App</i> Built an Android app using Java that detects driving state via accelerometer and GPS speed thresholds, automatically restricting phone usage above 20 km/h.
XTree <i>Web Technologies</i> Developed a Chrome extension using JavaScript that scrapes and exports all anchor tags from any webpage to a CSV file, reducing manual data collection from ~10 minutes to under 5 seconds per page.
PXP-Fraudnet <i>Machine Learning</i> - Architected a 3-layer feedforward neural network for fraud detection in payment transactions. - Designed a React.js dashboard displaying real-time fraud risk scores, transaction timelines, and model confidence metrics for analyst review.
Translator <i>Natural Language Processing</i> - Developed a text translation model to convert multilingual text into English. - Applied tokenization, stopwords removal, and sequence padding on multilingual corpora to preprocess input for a seq2seq translation model.

Work Experience

AI Solutions Engineer | Payintelli

June 2025 – Current

Next.js, AWS, Typescript, Python

- Built and maintained the production dashboard (app.payintelli.com) for an AI-driven payment orchestration platform, developing UI for real-time fraud risk scores, smart transaction routing, and automated payment insights.
- Developed the marketing site and product pages for 4 AI-powered fintech products (fraud detection, payment routing, reconciliation, conversational analytics) using Next.js and React.js.
- Worked cross-functionally with the AI/data team to accurately surface model-driven outputs (risk scores, routing decisions, predictive insights) in production-facing interfaces.

Intern | PXP

February 2025 – May 2025

React.js, PyTorch

- Constructed a PyTorch-based fraud detection model trained on 10,000+ transaction records, achieving 94% classification accuracy and a 25% increase in recall while maintaining under 1% FPR.
- Exposed ML model predictions via a REST API integrated with a React.js dashboard, enabling real-time transaction risk visualization for end users.
- Delivered a production-ready ML pipeline covering data preprocessing, model training, evaluation, and frontend integration within a 3-month internship.

Virtual Intern for Google AIML | EduSkills AICTE COHORT-8

April 2024 – June 2024

TensorFlow, OpenCV, Google Colab Notebook

- Completed 3 end-to-end ML projects over 8 weeks, including a 5-class image classification CNN with >90% validation accuracy and a YOLOv5 object detection model served via Flask API.

Member | MLRIT CIE

July 2022 – October 2023

Web Technologies, Figma, Photoshop

- Shipped 3+ frontend features for the MLRIT CIE website using Wordpress, HTML, CSS, and JavaScript, improving navigation and event visibility for 500+ student users.
- Mentored 100+ students in project development and technical fundamentals.
- Coordinated technical events including Inventron and Metaloop.

Certifications

Data Structures and Algorithms | Issued by SmartInterviews

Introduction to Machine Learning | Issued by NPTEL

Introduction to Artificial Intelligence | Issued by LinkedIn Learning

Python | Issued by HackerRank

Java | Issued by HackerRank

Languages

English – Fluent | Hindi – Fluent | Telugu – Native